ABSTRACT

Smart contracts are self-executing programs deployed on decentralized blockchains that manage billions of dollars in decentralized finance (DeFi). Due to blockchain immutability, post-deployment bugs cannot be patched retroactively, routinely resulting in massive and irreversible financial losses. Traditional auditing methods—such as static analysis and symbolic execution—struggle with high false positive rates and computational state space explosion. Conversely, while emerging Large Language Models (LLMs) show exceptional code comprehension capabilities, single-agent configurations suffer from cognitive hallucinations, bias accumulation, and degeneration-of-thought when evaluating complex contract logic. This seminar presents a comprehensive review of the research paper **"Advanced Smart Contract Vulnerability Detection via LLM-Powered Multi-Agent Systems"** by Z. Wei, J. Sun, Y. Sun, Y. Liu, D. Wu, et al., published in *IEEE Transactions on Software Engineering*, Vol. 51, No. 10, October 2025. The paper proposes **LLM-SmartAudit**, a novel framework that transforms smart contract auditing from solitary prompting into a collaborative, multi-agent conversational architecture.

The proposed LLM-SmartAudit framework organizes specialized LLM agents into distinct operational personas: a Project Manager (PM), a Smart Contract Auditor (SCA), a Solidity Programming Expert (SPE), and a Smart Contract Counselor (SCC). The framework coordinates two complementary inspection strategies: Broad Analysis (BA) mode, which leverages ReAct-based Thought-Reasoning to uncover undocumented security weaknesses across the entire contract logic, and Targeted Analysis (TA) mode, which employs Buffer-of-Thoughts (BoT) template reasoning across 40 predefined critical vulnerability classes. When candidate findings diverge, agents engage in a multi-turn collaborative debate with bidirectional role-swapping to eliminate confirmation bias before reaching consensus. Evaluated across 110 common vulnerability benchmark contracts, 6,454 real-world Code4rena contracts, and 13 critical CVEs, LLM-SmartAudit achieves 98% overall accuracy on common vulnerabilities, detects 12 out of 13 critical CVEs (surpassing PropertyGPT), and demonstrates an economical operational cost of ~$0.98 USD per audited contract. The framework provides interpretable, human-readable mitigation steps, presenting a robust and practical paradigm for automated smart contract security verification.

Keywords : Smart Contracts, Vulnerability Detection, Multi-Agent Systems, Large Language Models, LLM-SmartAudit, Decentralized Finance (DeFi) Security.

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